Alexander Robles

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Github: https://github.com/Alro10
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Google Scholar: 18 US patents (5 granted) in ML, GenAI and quantum computing

Experience

April 2026 - now **Tech AI Lead - Syngenta**.

- Leading GraphRAG development for CropwiseAI, Syngenta Digital's AI assistant built on top of other Syngenta Digital products, with focus on the fungicides, herbicides and insecticides product lines.

April 2026 - now **Founder - Rangoai**.

- Founder at Rangoai, building AI Agentic Systems for food topics.

September 2026 - now **Forward Deployed Engineer (part-time) - Slyng**.

- Delivering production AI agents and automated routines for an industrial client, with a multi-tenant architecture (organization-based isolation).
- Built the containerized deployment pipeline and secret manager used across staging and production environments.
- Developed a knowledge-agent that curates and serves domain context to the client's AI assistant.

December 2023 - April 2026 **Tech Lead GenAI Solutions - Meutudo**.

- Build and code a game changer solution: AI Assistant for customer support (24/7). This solution serves 90 percent of total interactions (200K per month) with 80 percent of CSAT. The solution is based on Multi-Agents and RAG with the following stack: vectorDB, SOTA OpenAI GPT and Claude models, prompt engineering and MCP.
- Ticket reason solution by using LLM model for classification, this impacts directly in customer team analysis objective that is create a better experience for customers.
- Building the first (in Brazil) whatsapp-app (based on Agents) solution where the customers can make loans (such as FGTS, Consignado Privado, INSS).

May 2022 - November 2023 **Senior Research Scientist - Dell Technologies**.

- Led initiatives in Generative AI and state-of-the-art Large Language Models (LLMs); 25 invention disclosures approved by Dell's patent committee, filed as US patents (see Patents).
- Explore the limits of reasoning of LLMs.
- Fine-tuning and serving Generative Models.
- Building the next game changer Generative AI platform for Dell.

May 2021 - May 2022 **Senior Data Scientist - IBM**.

- I led squads in Contract Life cycle Management tribe and guided the data science team to focus on delivering top-notch data science solutions for IBM contracts. We delivered two principal solutions for contract analysis: the first was a classification model to infer contract category and the second was a clustering model to catch possible errors in the contract repository to preserve data quality in the long term.

July 2019 - April 2021 **Data Scientist at IBM**.

- NLP for finding text similarity in documents. NLU for obtaining entities from queries to help the client extract useful information from the contract repository.
- Build a platform for CV analysis, match with job description and main skills. Save data and enable smart queries.
- Build APIs based on microservices that are deployed in Kubernetes with CI/CD. The top-notch solution needs to use innovative stacks such as Python, MongoDB, Kafka, ElasticSearch, Docker, Kubernetes.

Education

2018–2023 **PhD student in Computer Engineering**, *University of Campinas, FEEC*, Campinas-SP, Brazil, *Topics: Deep Learning, Time Series Forecasting and Machine Learning*.

2016–2018 **MSc in Electrical and Computer Engineering**, *University of Campinas, FEEC*, Campinas-SP, Brazil, *GPA 3.5/4.0*.

2007–2012 **BSc in Mechatronic Engineering**, *National University of Engineering (UNI)*, Lima-Peru.

Patents and Publications

- Granted US patents:
- US 12,585,532: Framework for selecting thresholds for anomaly detection models and generating quantitative explanations.
- US 12,676,875: Leveraging global explanations on a restricted edge environment.
- US 12,641,101: Association rules for root cause tracking and concise explanations in anomaly detection.
- US 12,555,562: Voice synthesis from diffusion generated spectrograms for accessibility.
- US 12,495,055: Non-anonymized privacy preserving global and local anomaly detection in distributed systems.
- Pending applications include LLM training estimation, natural language to QUBO code generation, quantum circuit knitting and federated learning. Peer-reviewed papers on time-varying system identification (PhD). Full list on Google Scholar.

Software skills and Certifications.

Programming Languages: Pytorch, Python and JavaScript.

Other tools and Clouds: SQL, PostgreSQL, MongoDB, Tableau, Docker, Kubernetes, git, gitflow, AWS, Azure.

Certifications: Deep Learning Specialization and Machine Learning Course by Andrew Ng. **To visualize my badges, please access to my Credly profile:** Credly

Languages

Fluent: Spanish, Portuguese and English